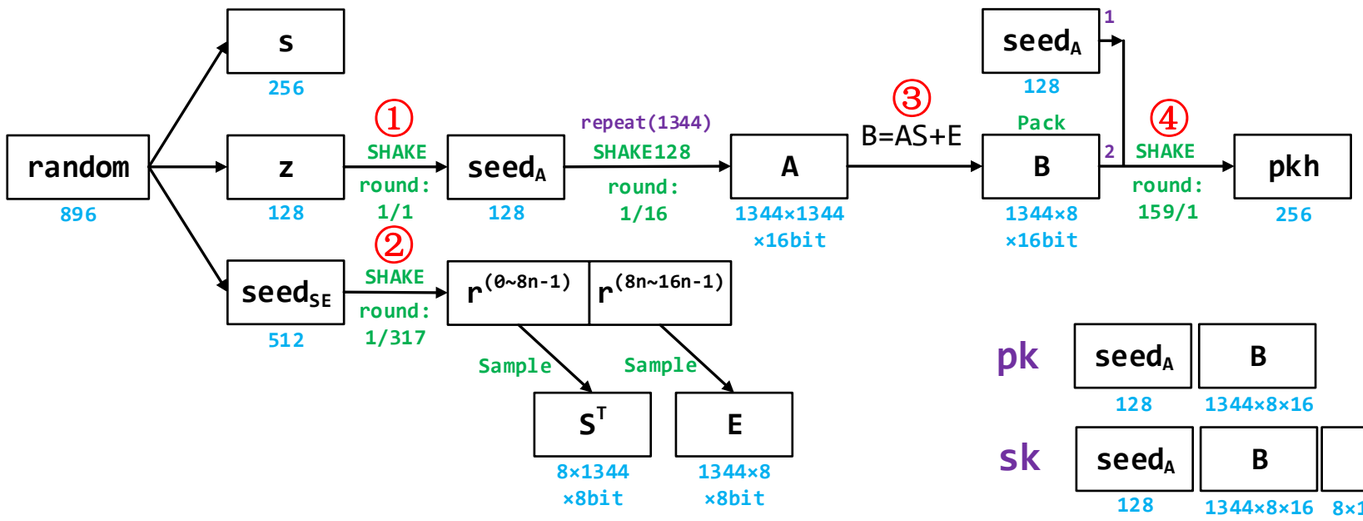
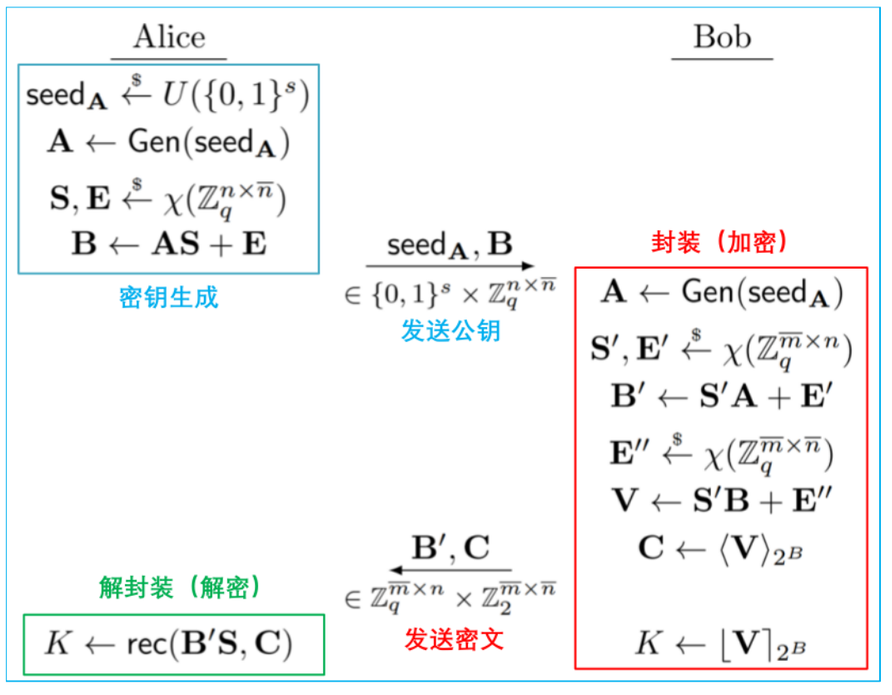
FrodoKEM-1344

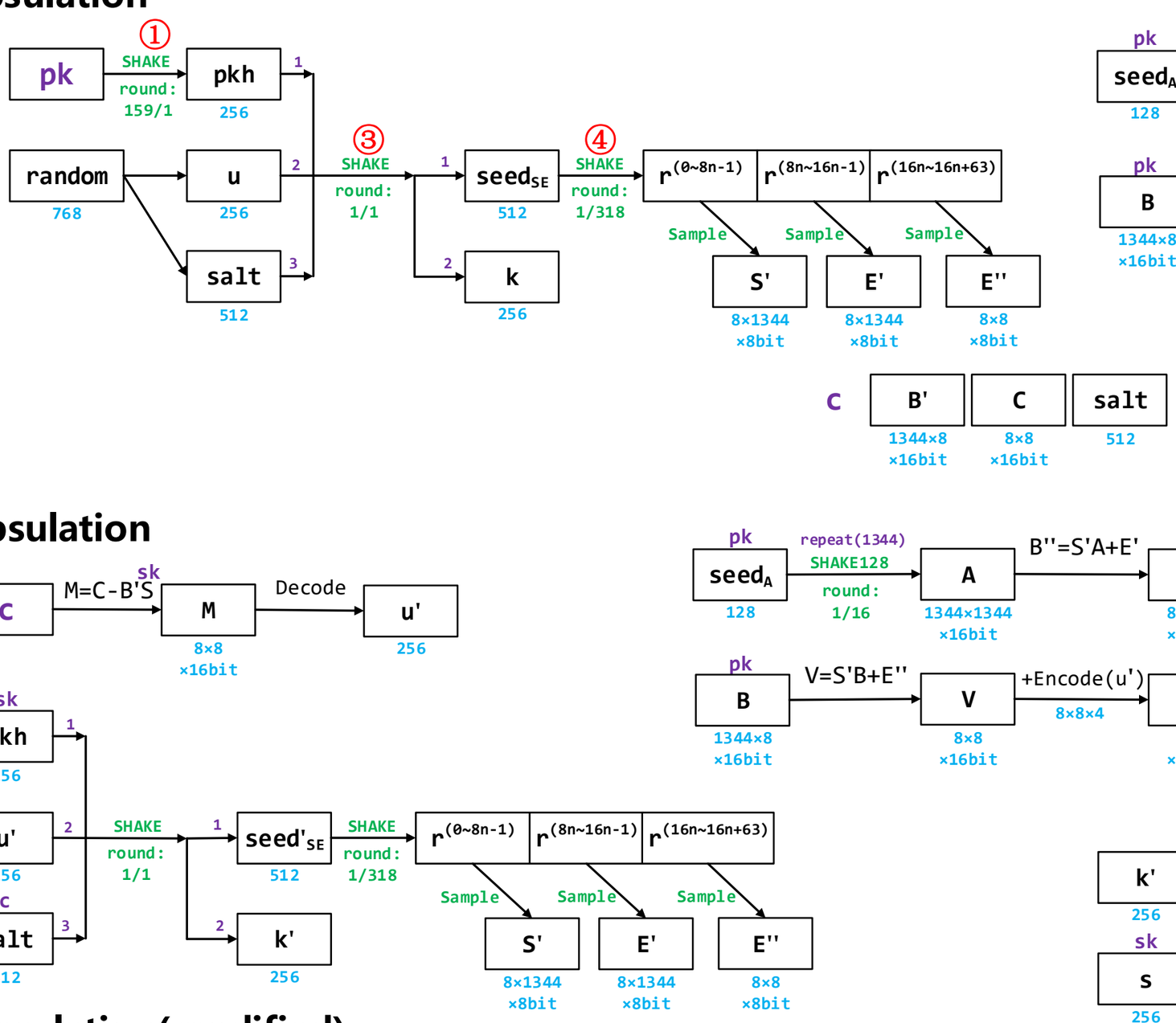
SHAKE
round:
absorb rounds/squeeze rounds

Key generation



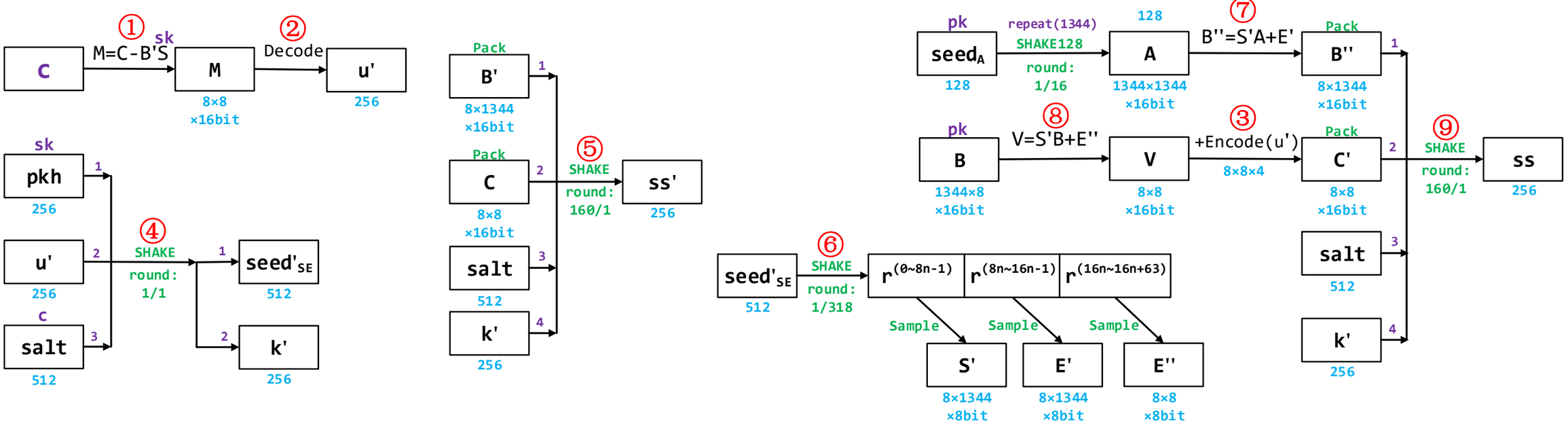


Encapsulation



Decapsulation

Decapsulation(modified)



循环方向(从内到外)
第1层
第2层
第3层

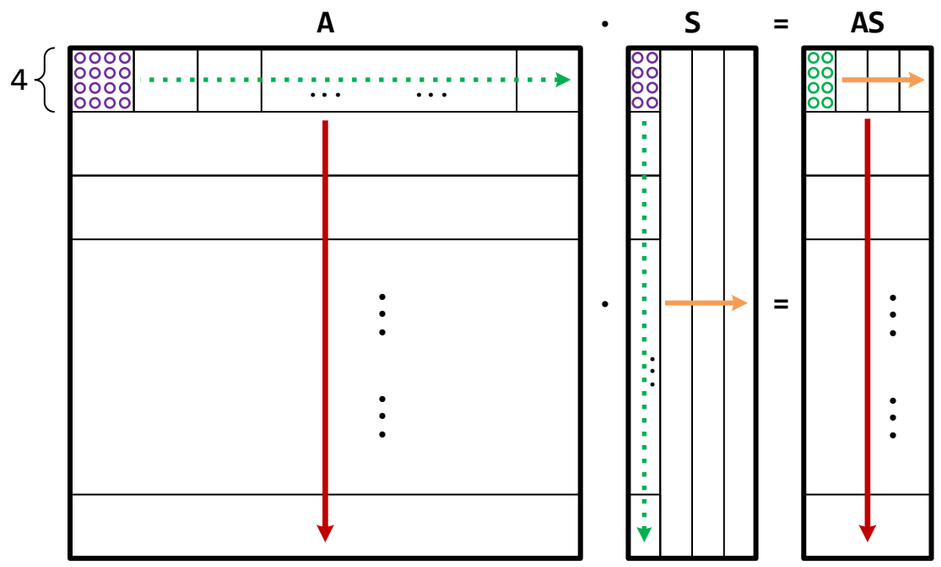
版本v4用到了两个4×4的乘法器阵列

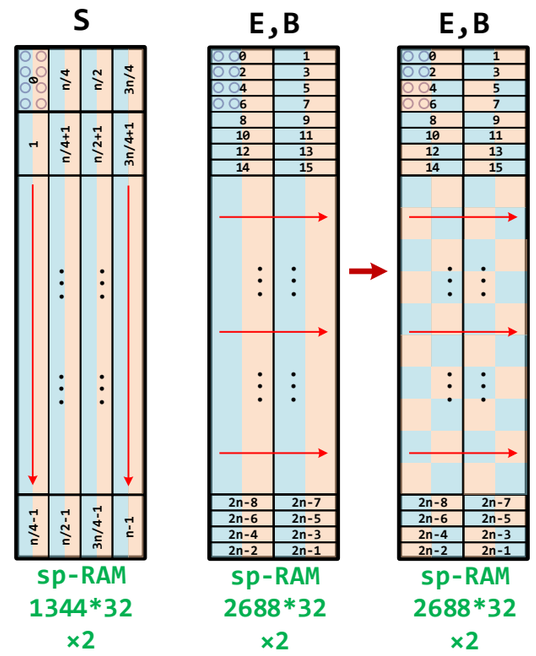
S,B sp-ram 4032*32 ×2
S',B' dp-ram 2016*32 ×4

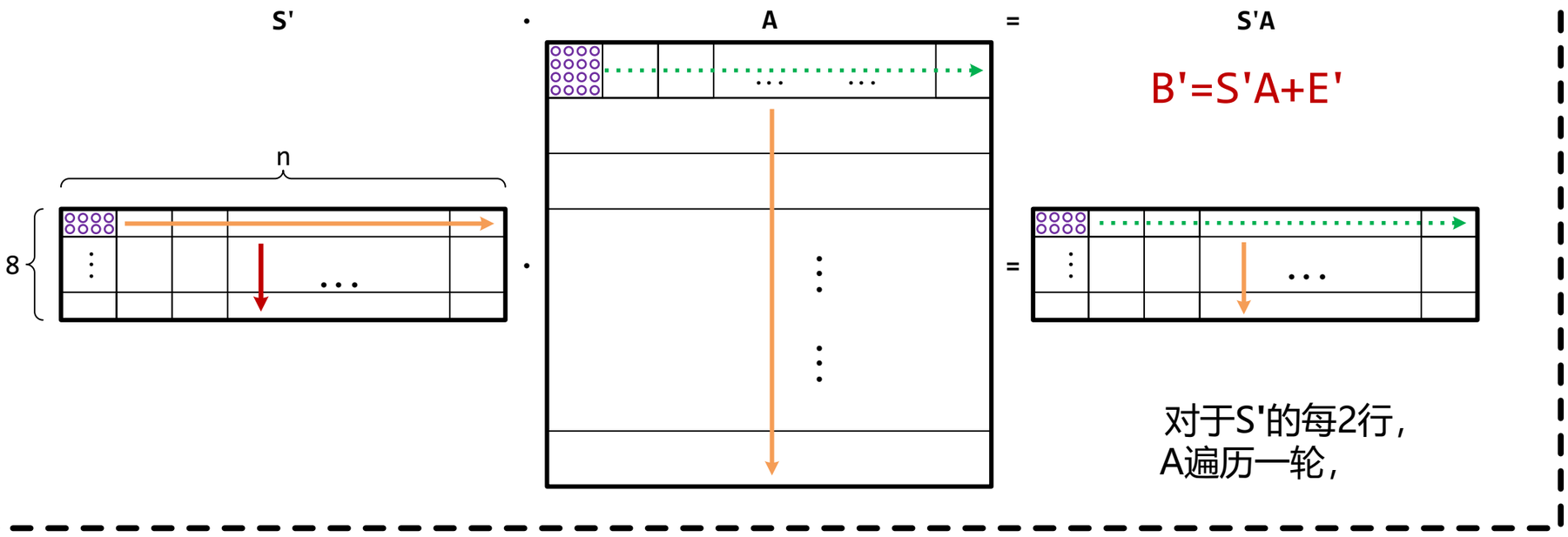
地址安排
RAM0
RAM1

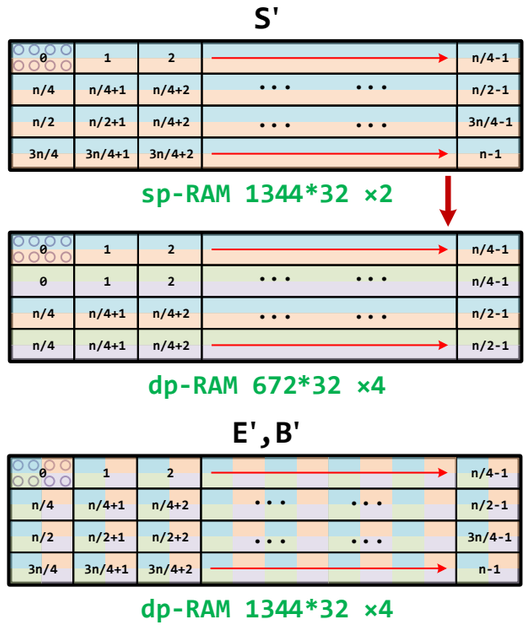
矩阵生成
S E

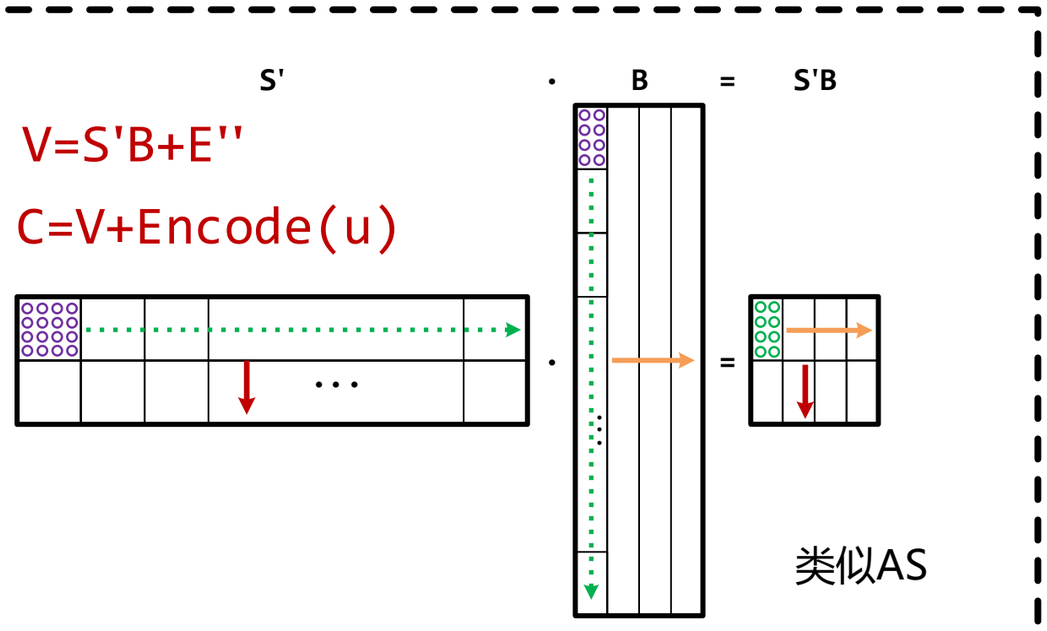
矩阵生成
S'
E'
E''

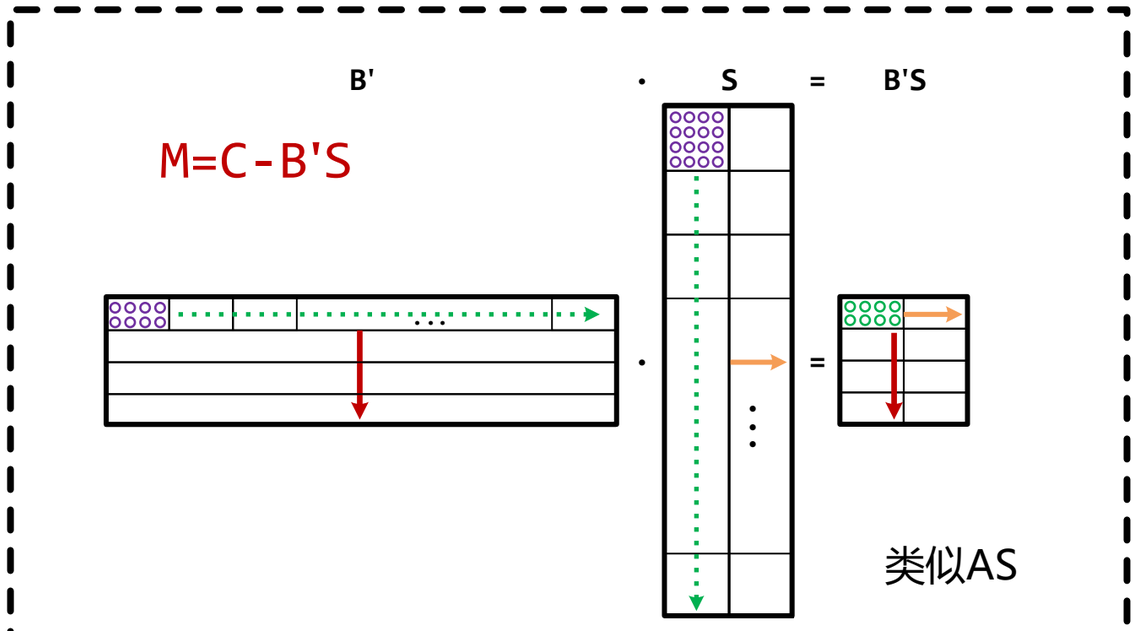








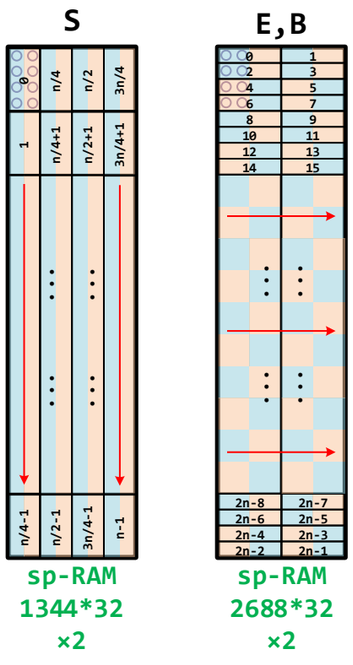


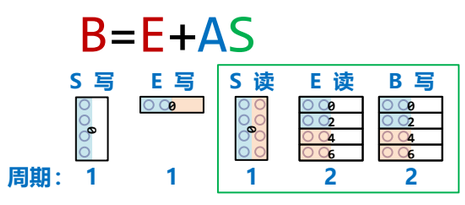


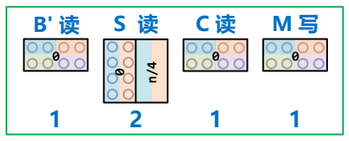
RAM0
RAM1
RAM2
RAM3

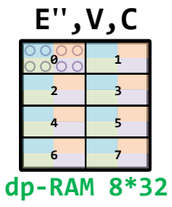
sp-ram 4032*32×2

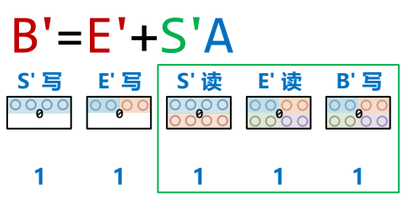
读写时序

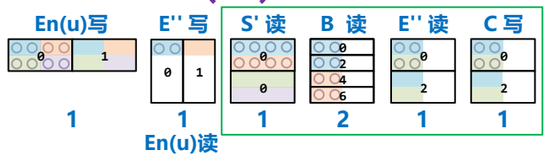




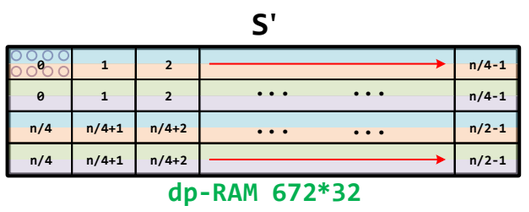


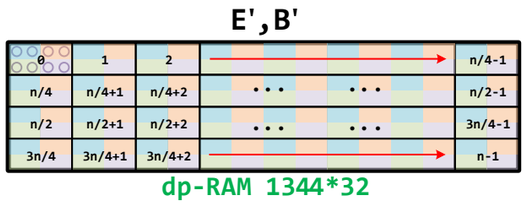




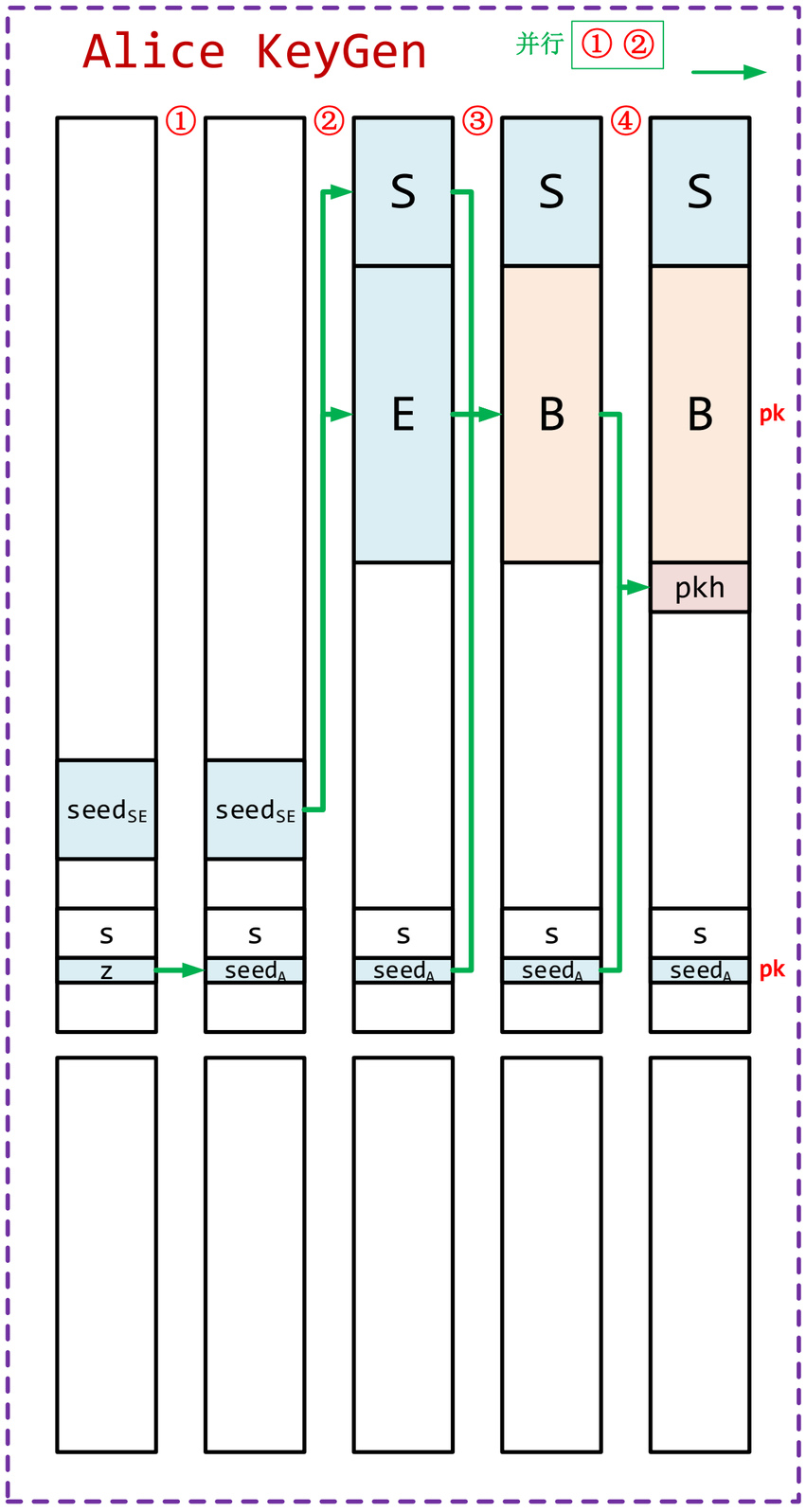


dp-ram 2024*32×4

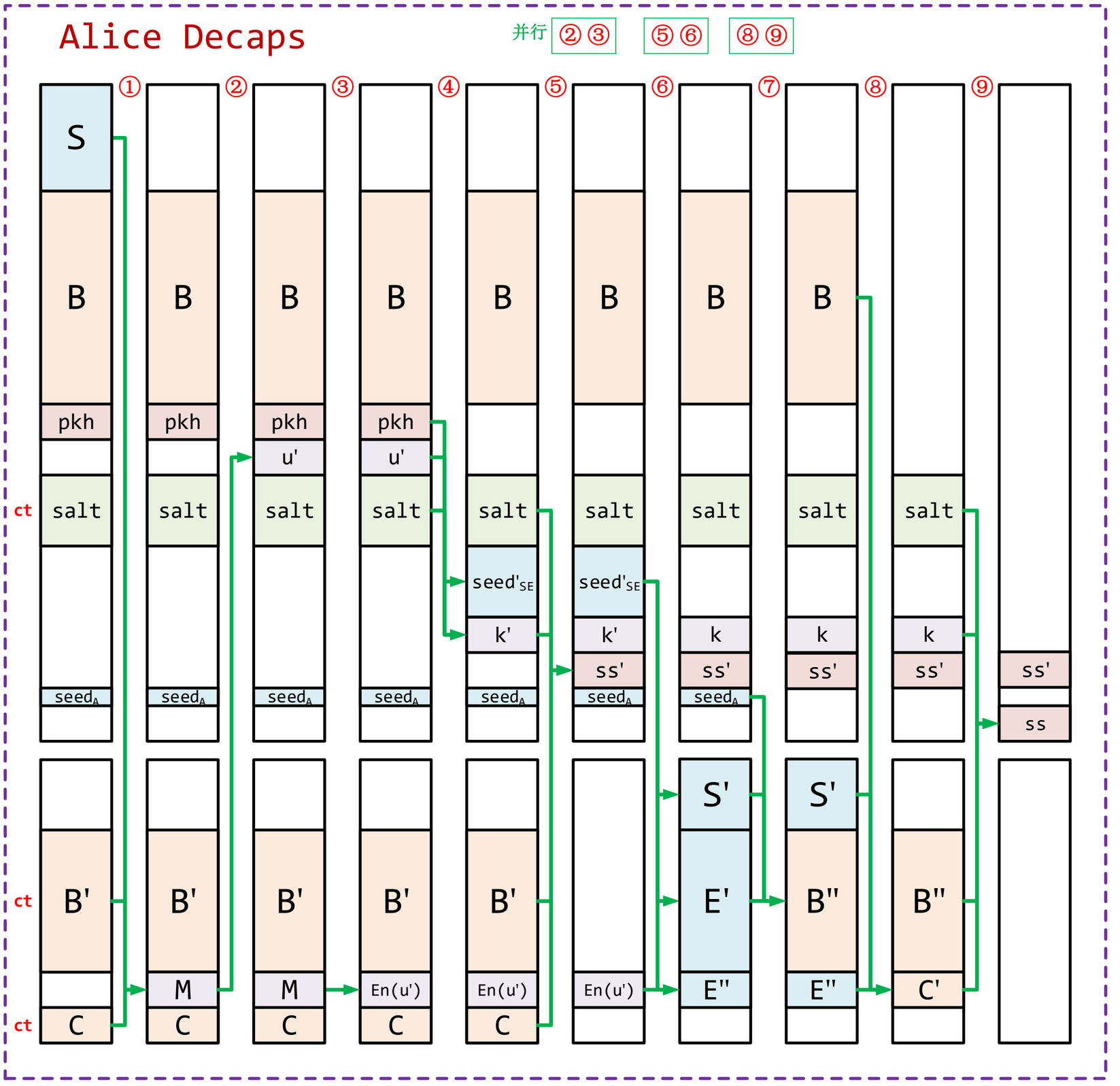


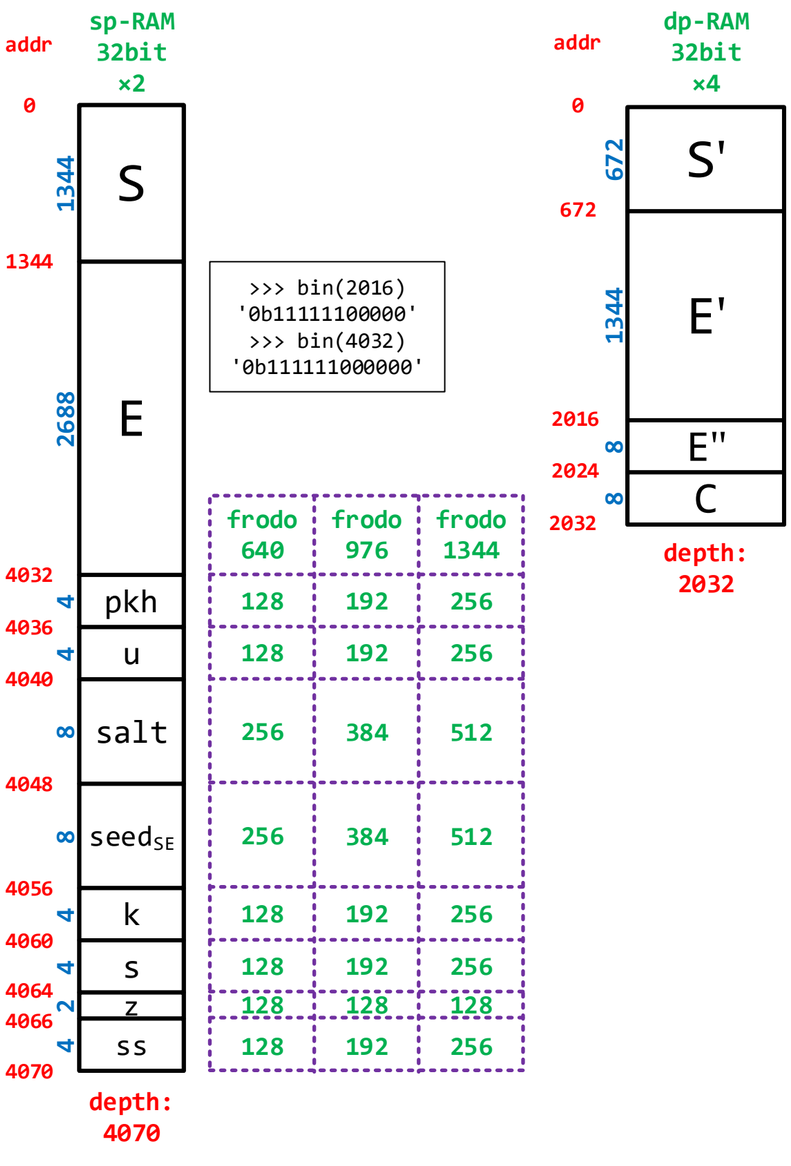


Alice KeyGen

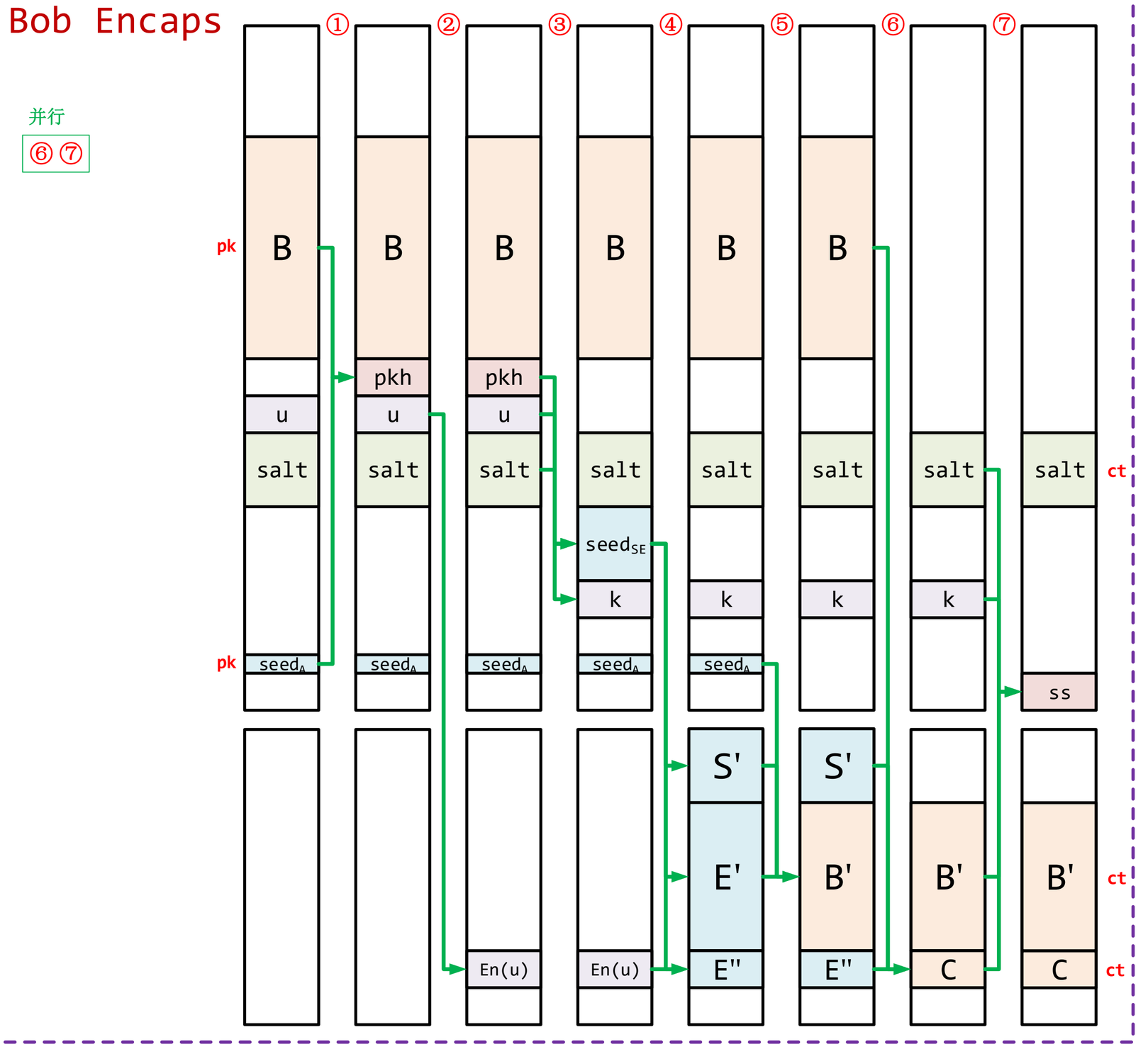


Alice Decaps





Bob Encaps



RAM width : 64
round: 吸收/挤压轮数
cycle: 读写RAM的周期

2024/8/26

t →

Key generation

SHAKE
reg 128bit
z → $seed_A$

round: 1/1

SHAKE
RAM 1344*64
生成S存入RAM

round: 1/159
cycle: 1344/4*8

(159*24=3816)

SHAKE
RAM 1344*128
生成E存入RAM

round: 1/159
cycle: 1344/4*8

SHAKE × 4
$seed_A$ → A
32个16*5乘法器
生成A，计算B

生成A的4行
1 | 2 | ... | 1344/4

按列遍历S
1 | 2 | ... | 1344/4

cycle: 1344/4*8

存B前4行
1 | 2 | ... | 最后4行 336

存B，
存到E的位置

SHAKE
生成pkh

round: 159/1
cycle: 1344/4*8

Encapsulation

SHAKE
生成pkh

round: 159/1
cycle: 1344/4*8

SHAKE
reg 512bit
生成$seed_{SE}$

round: 1/1

SHAKE
RAM 1344*64
生成S'存入RAM

round: 1/159
cycle: 1344/4*8

SHAKE
RAM 1344*128
生成E'存入RAM

round: 1/159
cycle: 1344/4*8

SHAKE
reg 8*8*5
生成E''存入reg
提前+Encode(u)

round: 1/1

多次遍历S'的前4列
1 | 2 | ... | 1344/4

cycle: 1344/4*8

边取边存B'
1 | 2 | ... | 最后1轮 336

取、累加、存，
存到E'的位置

SHAKE × 4
$seed_A$ → A
32个16*5乘法器
生成A，计算B'

生成A的4行
1 | 2 | ... | 最后4行 1344/4

32个16*5乘法器
计算C

B前4行
1 | ... | B前4行 1344/4

S'前4列 S'前4列
1 | 1 | 1344/4 | 1344/4

cycle:6

存C
1

SHAKE
生成ss

round: 160/1
cycle: 1344/4*8

336

Decapsulation

重复Decap部分

16个16*5乘法器
仿照计算B'S，计算M

S前4行
1 | ... | S前4行 1344/4

B'前4列 B'前4列
1 | 1 | 1344/4 | 1344/4

cycle:6

存M
1 | 336

计算u'

SHAKE
reg 512bit
生成$seed_{SE}$

round: 1/1

SHAKE
RAM 1344*64
生成S'存入RAM

round: 1/159
cycle: 1344/4*8

SHAKE
RAM 1344*128
生成E'存入RAM

round: 1/159
cycle: 1344/4*8

SHAKE
reg 8*8*5
生成E''存入reg
提前+Encode(u)

round: 1/1

SHAKE × 4
$seed_A$ → A
32个16*5乘法器
生成A，计算B'

生成A的4行
1 | 2 | ... | 最后4行 1344/4

多次遍历S'的前4列
1 | 2 | ... | 1344/4

cycle: 1344/4*8

边取边存B'
1 | 2 | ... | 最后1轮 336

取、累加、存，
存到E'的位置

32个16*5乘法器
计算C

B前4行
1 | ... | B前4行 1344/4

S'前4列 S'前4列
1 | 1 | 1344/4 | 1344/4

cycle:6

存C
1

SHAKE
生成ss

round: 160/1
cycle: 1344/4*8

336